

PHARMACOLOGICAL STUDIES OF ADIANTUM CAPILLUS - VENERIS

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ABSTRACT

An aqueous and a petrolether extract of the medicinal plant *Adiantum capillus - veneris* were prepared and their effects on diuresis and sodium and potassium level in urine were studied on guinea pigs and compared with the effect of furosemide. Both extracts affects the studied parameters, however the petrolether extract seems to be more potent than the aqueous. By comparing the results of *Adiantum* extracts and furosemide with the control we can assume a similar mechanism of action.

INTRODUCTION

Herbal medicine has been and still is widely-spread practice in Iraq. *Adiantum* is one of important local plants which is of value for medicinal purposes. Tradition has attributed to various species of this genus of ferns valuable properties in chronic pulmonic catarrhs in Europa and in India.

The European species was formerly employed as emmenagogue - under the names of *polytrichi*, *polytrichon* or *kalliphyllon* given in the form of infusion, sweetened with sugar or honey and a syrup prepared from it (*syrops Adianti*, *syrops capilli veneris*) has official use in a number of european pharmacopias (1943).

We undertook pharmacological investigation of *Adiantum* Iraqi indigenous Iraqi medicinal plant. Researchers (Al-Rawi & Chakravarty 1964 & Chakravarty 1976) reviewed the uses in folk medicine of hundred of botanically identified Iraqi plants. Their lists, together with reports from local herbalists, served as a working guide in the diuretic and other pharmacological studies.

This study was undertaken to compare the diuretic activity of *Adiantum capillus - veneris* (Fig. 1) with frusemide to see if there is a promising effect of *Adiantum* extract in management of stone or renal disorders.

MATERIALS AND METHODS

Plant material:

The plant was collected from north region of Iraq near Sulaymaniah district and dried, *Adiantum* was authenticated by the staff of the Iraqi national herbarium in Abu-Ghraib.

1. Preparation of extracts (Ext A):

Freshly collected dried plant material (400 g) was ground into coarse powder and extracted at room temperature with 50% petroleum ether (1000 ml of each). The solvent was evaporated in vacuum at 20c to give the drycrude extract. The extracts were dissolved in enough physiological saline solution to give the required concentration. 1 gm of dry freshly prepared extract dissolved in 10cc of saline as needed for pet. ether and aqueous extract.

2. Aqueous extract (Ext B):

Plant material (400 g) was macerated in a blender and boiled with water (2 litre) for 1 hour. The extract was filtered and the residue extracted in a similar way. The filtrates were combined and evaporated under reduced pressure to 500 ml.

Drugs:

Lasix (Hoechst), Histamine acid phosphate (BDH), Acetylcholine (Fluka), Atropine sulphate monohydrate (Fluka) Diphenhydramine (SDI), Nicotine (Hopkin & Williams).

Experimental Animals:

Guinea-pigs (350-450 g) were utilized. The animals were allowed food and water ad libitum before and throughout the experiments.

Diuretic Test:

For each test 12 animals were placed in metabolic cages. The animals were divided into four groups, group (1) served as control without any drug or extract, group (2) received drug extract intraperitoneally, group (3) received saline intraperitoneally, group (4) received frusemide 35mg/kg intraperitoneally (see table 1). The test dose of extract under study: doses are expressed in mg of the extract.

Urine was collected over a period of 24 h. The experiment was repeated four times for each group.

Isolated Guinea-Pig ileum:

Guinea-pigs were killed by stunning and exsanguination by sectioning of the carotid arteries. A piece of ileum was removed and cut into 3 cm long sections and placed in Tyrode solution maintained at 37 c. The pieces were transferred to a 50 ml organ bath containing Tyrode solution maintained at 37 c and gassed continuously with a mixture of 95% O₂ + 5% CO₂. Contractions of there ileum were recorded on a kymograph.

STATISTICAL ANALYSIS

Statistical significance was calculated following the assessment method adopted (Cum-mings et al., 1960). The criteria for acceptance or rejection in both directions (diuresis and antidiuresis were; for the 24h test, a change of 75% or more in both directions in the amount

of urine caused by the extracts as compared to the control was accepted as positive or significant. A change of 15% or less was considered insignificant.

RESULTS AND DISCUSSION

Results depicted in table 1 and 2 are the means $\pm$ SEM of 6 experiments. *Adiantum capillus* produced positive diuresis in the 24 h test (Twaij et al. 1985) compared with furosemide. However *Adiantum* extract did not show any contracting or relaxing effect of the isolated ileum but this extract antagonizes the contracting effect of nicotine (Fig. 2). The *Adiantum* extract did not antagonize the acetylcholine and histamine effect on isolated guinea-pig ileum. This may suggest that this extract may have ganglionic-blocking activity (Sharif 1976). *Adiantum capillus - veneris* has shown a promising diuretic activity (1987). However work is going on first to identify the active constituent with such diuretic activity to elucidate the diuresis phenomena.

Table 1
This table shows the diuretic comparative effect between the effect of *Adiantum* and furosemide

Treatment	Urine vol. in ml	pH	Na ⁺ mmol/ml	K ⁺ mmol/ml
Control	12 $\pm$ 2	8 $\pm$ 2.5	85.67 $\pm$ 3.53	2.37 $\pm$ 0.06
<i>Adiantum</i> 10mg/kg (aqueous extract)	39 $\pm$ 3*	9 $\pm$ 2	80.00 $\pm$ 11.42	2.47 $\pm$ 0.30
Furosemide 2 mg/kg	38 $\pm$ 4*	9 $\pm$ 1	76.30 $\pm$ 6.29	2.85 $\pm$ 0.09**

Results are the means $\pm$ SEM of experiments. The experiments were performed on fed guinea-pigs each group of 3 animals.

*p < 0.05 ** p < 0.001

Table 2

This table shows the diuretic comparative effect between the effect of adiantum and furosemide

Treatment	Urine vol. in ml	pH	Na ⁺ mmol/ml	K ⁺ mmol/ml
Control	12 ± 2.5	8 ± 2	85.67 ± 3.53	2.37 ± 0.06
Adiantum 10mg/kg (aqueous extract)	40 ± 3*	9 ± 4	74.30 ± 15.55	2.38 ± 0.33
Furosemide 2 mg/kg	38 ± 3*	9 ± 1	76.83 ± 6.29	2.85 ± 0.09**

Results are the means ± SEM of experiments. The experiments were performed on fed guinea-pigs each group of 3 animals.

*p < 0.05

**p < 0.001



Fig. 1: Typical picture of the plant *Adiantum capillus - veneris*.

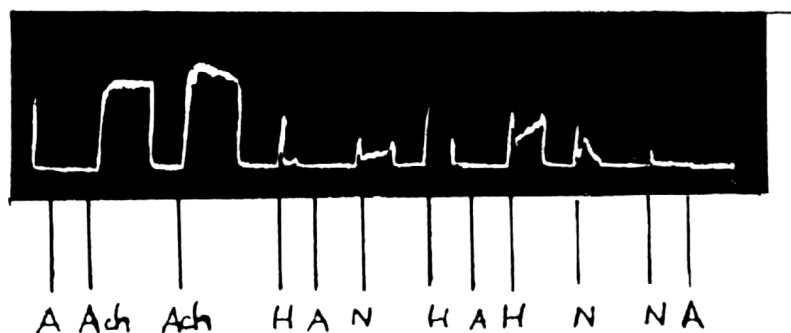


Fig. 2: Typical trace of isolated guinea-pig ileum showing the antagonistic effect of nicotine to adiantum. Adiantum extract (A) 200 mg/ml, Acetylcholine (Ach) 10^{-6} m Histamine (H) 2×10^{-5} m, Nicotine (N) 4×10^{-4} m

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